

NADRA-CLAIRE ASSAD

Houston, Texas

☎ 832-819-8806 ✉ nadraclairea@gmail.com 💻 [linkedin.com/in/nadraclairea/](https://www.linkedin.com/in/nadraclairea/)
U.S. Citizen — Authorized to work in the United States without sponsorship

Education

University of Houston

Aug. 2023–May 2027

Bachelor of Science in Computer Engineering

Houston, Texas

- Academic Excellence Scholarship

Relevant Coursework

- Automatic Control Systems
- Analog Electronics
- Applied Electromagnetics
- Signals and Systems
- Microprocessor Systems
- Digital Logic Design

Experience

Children's Museum Houston | *Soldering, TinkerCAD, 3D printing*

May–August 2024

Summer STEM Intern

Houston, Texas

- Assisted in leading electronics and 3D-printing workshops requiring soldering PCB boards and debugging Arduinos.
- Built a physics exhibit for ExxonMobil's MindFest in which a large lemon-powered battery array powered calculators, clocks, and LEDs.
- Proposed new electricity and magnetism activities and researched devices for future STEM workshops.

Projects

PID Control for Servo Motor Position and Speed | *MATLAB, Simulink*

April–May 2026

- Modeled two DC servo motor systems in MATLAB/Simulink using transfer functions to simulate a linear actuator and rotating slotted disc.
- Designed and tuned closed-loop PID controllers to coordinate actuator position with disc angular motion for collision-free pass-through.
- Verified system performance through position and velocity plots, confirming the actuator stayed below its 4 m/s maximum speed limit.

USB-A LED Keychain PCB | *KiCad, PCB Layout*

Sep 2025

- Designed a compact PCB with a USB-A connector powering an onboard LED, including schematic capture, footprint assignment, and basic trace routing in KiCad.

Three Octave Electric Keyboard with Metronome and Note Display | *Arduino, ESP32, UART*

Sep–Nov 2025

- Designed and built a three-octave keyboard with accurate tone generation, octave switching, and RGB TFT screen note display.
- Implemented sustain and soften pedal functions, adjustable volume control, and precise 0.5 s/0.25 s note timing.
- Created a BPM-adjustable metronome with a potentiometer interface and integrated real-time BPM display.
- Implemented a 115200-baud UART interface between Arduino and ESP32-S3, with voltage level shifting and real-time note data rendering on the TFT display.

Impact of Extreme Temperature Cooling on Semiconductor Wavelength Emission | *Arduino*

Mar–May 2023

- Developed and presented a poster on semiconductor wavelength emission to Harvard and MIT PhD students.
- Built and operated a custom Arduino-driven red-laser setup and submerged LEDs in liquid nitrogen for testing.
- Analyzed wavelength shifts, observing major color changes in warm-tone LEDs while cool-tone LEDs remained stable.

Technical Skills

Programming: Assembly, C++, LaTeX

Hardware/Tools: Arduino, Oscilloscopes, DMMs, Soldering, UART Protocol

Software: MATLAB, Simulink, LTspice, KiCad, TinkerCAD

Extracurricular

Society of Women Engineers (SWE)

Fall 2023–Present

Project Chair

University of Houston

Institute of Electrical and Electronics Engineers (IEEE)

Fall 2023–Present

Member

University of Houston